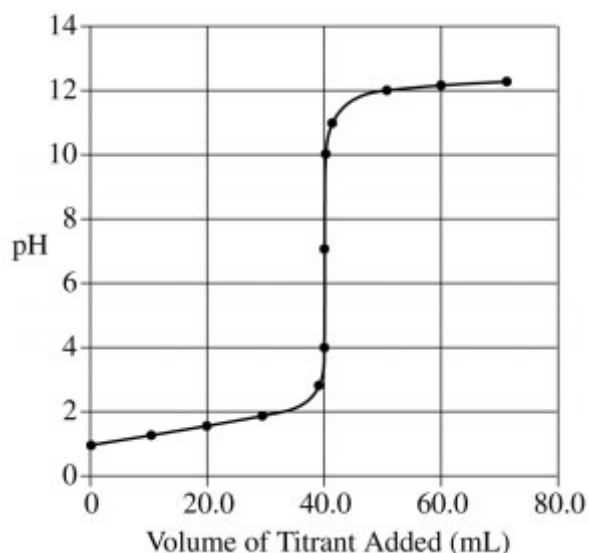


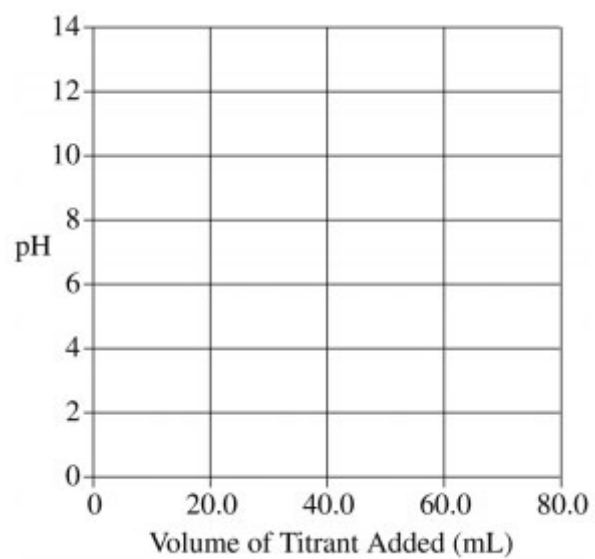
FR Practice #1 (2010B #5, 10 points)



1. A solution of 0.100 M HCl and a solution of 0.100 M NaOH are prepared. A 40.0 mL sample of one of the solutions is added to a beaker and then titrated with the other solution. A pH electrode is used to obtain the data that are plotted in the titration curve shown above.
 - a. Write the net ionic equation for the reaction of HCl(aq) and NaOH(aq). (1 point)
 - b. Identify the solution that was initially added to the beaker. Explain your reasoning. (1 point)
 - c. On the titration curve above, circle the point that corresponds to the equivalence point. (1 point)
 - d. At the equivalence point, how many moles of titrant have been added? (1 point)
 - e. The same titration is to be performed again, this time using an indicator. Use the information in the table below to select the best indicator for the titration. Explain your choice. (2 points)

Indicator	pH Range of Color Change
Methyl violet	0 – 1.6
Methyl red	4 – 6
Alizarin yellow	10 – 12

- f. What is the difference between the equivalence point of a titration and the end point of a titration? (1 point)
- g. On the grid provided on the next page, sketch the titration curve that would result if the solutions in the beaker and buret were reversed (i.e., if 40.0 mL of the solution used in the buret in the previous titration were titrated with the solution that was in the beaker). (3 points)



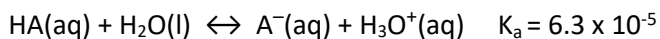
FR Practice #2 (2012 #1, shortened, 7 points)

2. A 1.22 g sample of a pure monoprotic acid, HA, was dissolved in distilled water. The HA solution was then titrated with 0.250 M NaOH. The pH was measured throughout the titration, and the equivalence point was reached when 40.0 mL of the NaOH solution had been added. The data from the titration are recorded in the table below.

Volume of 0.250 M NaOH Added (mL)	pH of Titrated Solution
0.00	?
10.0	3.72
20.0	4.20
30.0	?
40.0	8.62
50.0	12.40

- Explain how the data in the table above provide evidence that HA is a weak acid rather than a strong acid. (1 point)
- Write the balanced net-ionic equation for the reaction that occurs when the solution of NaOH is added to the solution of HA. (1 point)
- Calculate the number of moles of HA that were titrated. (1 point)
- Calculate the molar mass of HA. (1 point)

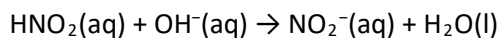
The equation for the dissociation reaction of HA in water is shown below.



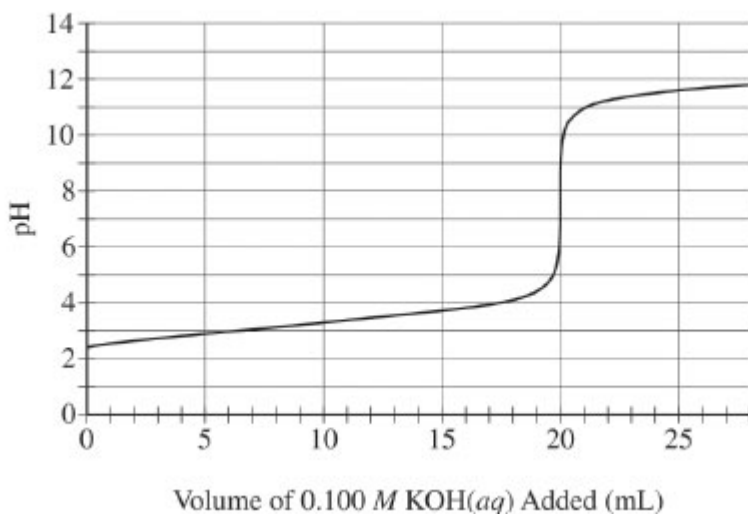
- Assume that the initial concentration of the HA solution (before any NaOH solution was added) is 0.200 M. Determine the pH of the initial HA solution. (3 points)

FR Practice #3 (2018 #2, shortened, 3 points)

3. To produce an aqueous solution of HNO_2 , a student bubbles $\text{N}_2\text{O}_3(\text{g})$ into distilled water. Assume that the reaction goes to completion and that HNO_2 is the only species produced. To determine the concentration of $\text{HNO}_2(\text{aq})$ in the resulting solution, the student titrates a 100. mL sample of the solution with 0.100 M $\text{KOH}(\text{aq})$. The neutralization reaction is represented below.



The following titration curve shows the change in pH of the solution during the titration.



- a. Use the titration curve and the information above to
- determine the initial concentration of the $\text{HNO}_2(\text{aq})$ solution (1 point)
 - estimate the value of the pK_a for $\text{HNO}_2(\text{aq})$ (1 point)
- b. During the titration, after a volume of 15 mL of 0.100 M $\text{KOH}(\text{aq})$ has been added, which species, $\text{HNO}_2(\text{aq})$ or $\text{NO}_2^-(\text{aq})$, is present at a higher concentration in the solution? Justify your answer. (1 point)

Quiz Free Response Practice (2009B, #1, 11 points)

A pure 14.85 g sample of the weak base ethylamine, $\text{C}_2\text{H}_5\text{NH}_2$, is dissolved in enough distilled water to make 500. mL of solution.

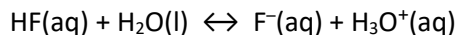
- (a) Calculate the molar concentration of the $\text{C}_2\text{H}_5\text{NH}_2$ in the solution. (2 points)
- (b) Write an equation showing how the aqueous ethylamine reacts with water. (1 point)
- (c) Write the equilibrium-constant expression for the reaction between $\text{C}_2\text{H}_5\text{NH}_2(\text{aq})$ and water. (1 point)
- (d) Of $\text{C}_2\text{H}_5\text{NH}_2(\text{aq})$ and $\text{C}_2\text{H}_5\text{NH}_3^+(\text{aq})$, which is present in the solution at the higher concentration at equilibrium? Justify your answer. (1 point)

- Calculate the concentration of $\text{OH}^-(\text{aq})$ in the solution. (1 point)
- Write the net-ionic equation that represents the reaction that occurs when the $\text{C}_2\text{H}_5\text{NH}_2$ solution is mixed with the HCl solution. (1 point)
- Calculate the molar concentration of the $\text{C}_2\text{H}_5\text{NH}_3^+(\text{aq})$ that is formed in the reaction. (2 points)
- Calculate the value of K_b for $\text{C}_2\text{H}_5\text{NH}_2$. (2 points)

AP Free Response Practice #1 (2017 #3, shortened, 5 points)

1. Nitrogen monoxide, NO(g) , can undergo reactions to produce acids such as HNO_2 , a weak acid with a K_a of 4.0×10^{-4} and a $\text{p}K_a$ of 3.40.
 - a. A student is asked to make a buffer solution with a pH of 3.40 by using 0.100 M $\text{HNO}_2(\text{aq})$ and 0.100 M $\text{NaOH}(\text{aq})$.
 - i. Explain why the addition of 0.100 M $\text{NaOH}(\text{aq})$ to 0.100 M $\text{HNO}_2(\text{aq})$ can result in the formation of a buffer solution. Include the net ionic equation for the reaction that occurs when the student adds the $\text{NaOH}(\text{aq})$ to the $\text{HNO}_2(\text{aq})$. (2 points)
 - ii. Determine the volume, in mL, of 0.100 M $\text{NaOH}(\text{aq})$ the student should add to 100. mL of 0.100 M $\text{HNO}_2(\text{aq})$ to make a buffer solution with a pH of 3.40. Justify your answer. (2 points)
 - b. A second student makes a buffer by dissolving 0.100 mol of $\text{NaNO}_2(\text{s})$ in 100. mL of 1.00 M $\text{HNO}_2(\text{aq})$. Which is more resistant to changes in pH when a strong acid or a strong base is added, the buffer made by the second student or the buffer made by the first student in part (a)? Justify your answer. (1 point)

FR Practice #2 (2018 #5, 4 points)



2. The ionization of HF(aq) in water is represented by the equation above. In a 0.0350 M HF(aq) solution, the percent ionization of HF is 13.0 percent.
- a. Two particulate representations of the ionization of HF molecules in the 0.0350 M HF(aq) solution are shown below in Figure 1 and Figure 2. Water molecules are not shown. Explain why the representation of ionization of HF molecules in water in Figure 1 is more accurate than the representation in Figure 2. (1 point)
- (The key below identifies the particles in the representations.)

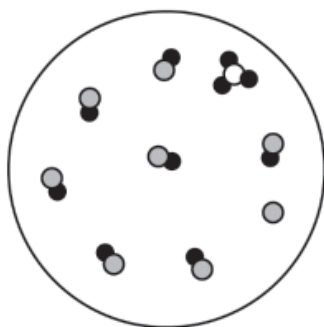
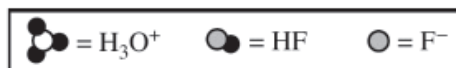


Figure 1

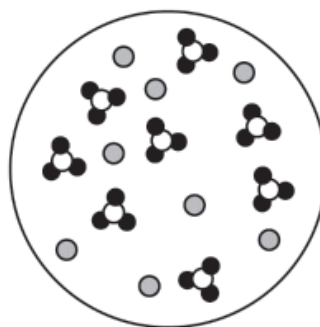
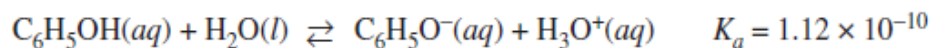


Figure 2

- b. Use the percent ionization data above to calculate the value of K_a for HF. (2 points)
- c. If 50.0 mL of distilled water is added to 50.0 mL of 0.0350 M HF(aq), will the percent ionization of HF(aq) in the solution increase, decrease, or remain the same? Justify your answer with an explanation or a calculation. (1 point)

FR Practice #3 (2016 #4, 4 points)



3. Phenol is a weak acid that partially dissociates in water according to the equation above.

a. What is the pH of a 0.75 M $\text{C}_6\text{H}_5\text{OH}(aq)$ solution? (2 points)

b. For a certain reaction involving $\text{C}_6\text{H}_5\text{OH}(aq)$ to proceed at a significant rate, the phenol must be primarily in its deprotonated form, $\text{C}_6\text{H}_5\text{O}^-(aq)$. In order to ensure that the $\text{C}_6\text{H}_5\text{OH}(aq)$ is deprotonated, the reaction must be conducted in a buffered solution. On the number scale below, circle each pH for which more than 50 percent of the phenol molecules are in the deprotonated form, $\text{C}_6\text{H}_5\text{O}^-(aq)$. Justify your answer. (2 points)

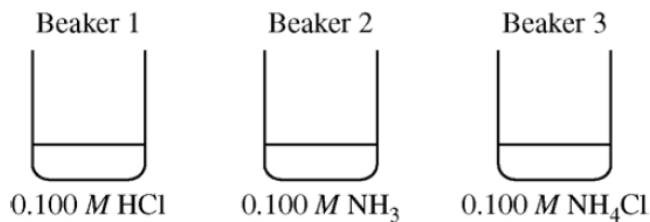
1 2 3 4 5 6 7 8 9 10 11 12 13 14

Free Response Practice #4 (5 points)

A pure 14.85 g sample of the weak base ethylamine, $\text{C}_2\text{H}_5\text{NH}_2$, is dissolved in enough distilled water to make 500. mL of solution.

- (a) Calculate the molar concentration of the $\text{C}_2\text{H}_5\text{NH}_2$ in the solution. (2 points)
- (b) Write an equation showing how the aqueous ethylamine reacts with water. (1 point)
- (c) Write the equilibrium-constant expression for the reaction between $\text{C}_2\text{H}_5\text{NH}_2(\text{aq})$ and water. (1 point)
- (d) Of $\text{C}_2\text{H}_5\text{NH}_2(\text{aq})$ and $\text{C}_2\text{H}_5\text{NH}_3^+(\text{aq})$, which is present in the solution at the higher concentration at equilibrium? Justify your answer. (1 point)

Free Response Practice #5 (2011 #1, 10 points)



2. Each of three beakers contains 25.0 mL of a 0.100 M solution of HCl, NH₃, or NH₄Cl, as shown above. Each solution is at 25°C.

a. Determine the pH of the solution in beaker 1. Justify your answer. (1 point)

b. In beaker 2, the reaction $\text{NH}_3(aq) + \text{H}_2\text{O}(l) \leftrightarrow \text{NH}_4^+(aq) + \text{OH}^-(aq)$ occurs. The value of K_b for $\text{NH}_3(aq)$ is 1.8×10^{-5} at 25°C.

i. Write the K_b expression for the reaction of $\text{NH}_3(aq)$ with $\text{H}_2\text{O}(l)$. (1 point)

ii. Calculate the $[\text{OH}^-]$ in the solution in beaker 2. (2 points)

- c. In beaker 3, the reaction $\text{NH}_4^+(aq) + \text{H}_2\text{O}(l) \leftrightarrow \text{NH}_3(aq) + \text{H}_3\text{O}^+(aq)$ occurs.
- Calculate the value of K_a for $\text{NH}_4^+(aq)$ at 25°C . (1 point)
 - The contents of beaker 2 are poured into beaker 3 and the resulting solution is stirred. Assume that volumes are additive. Calculate the pH of the resulting solution. (2 points)
- d. The contents of beaker 1 are poured into the solution made in part (c)(ii). The resulting solution is stirred. Assume that volumes are additive. Is the resulting solution an effective buffer? Justify your answer. (1 point)